



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

DIGITAL FORENSICS

System Forensics, Investigation, and Response

Lesson 10 Macintosh Forensics

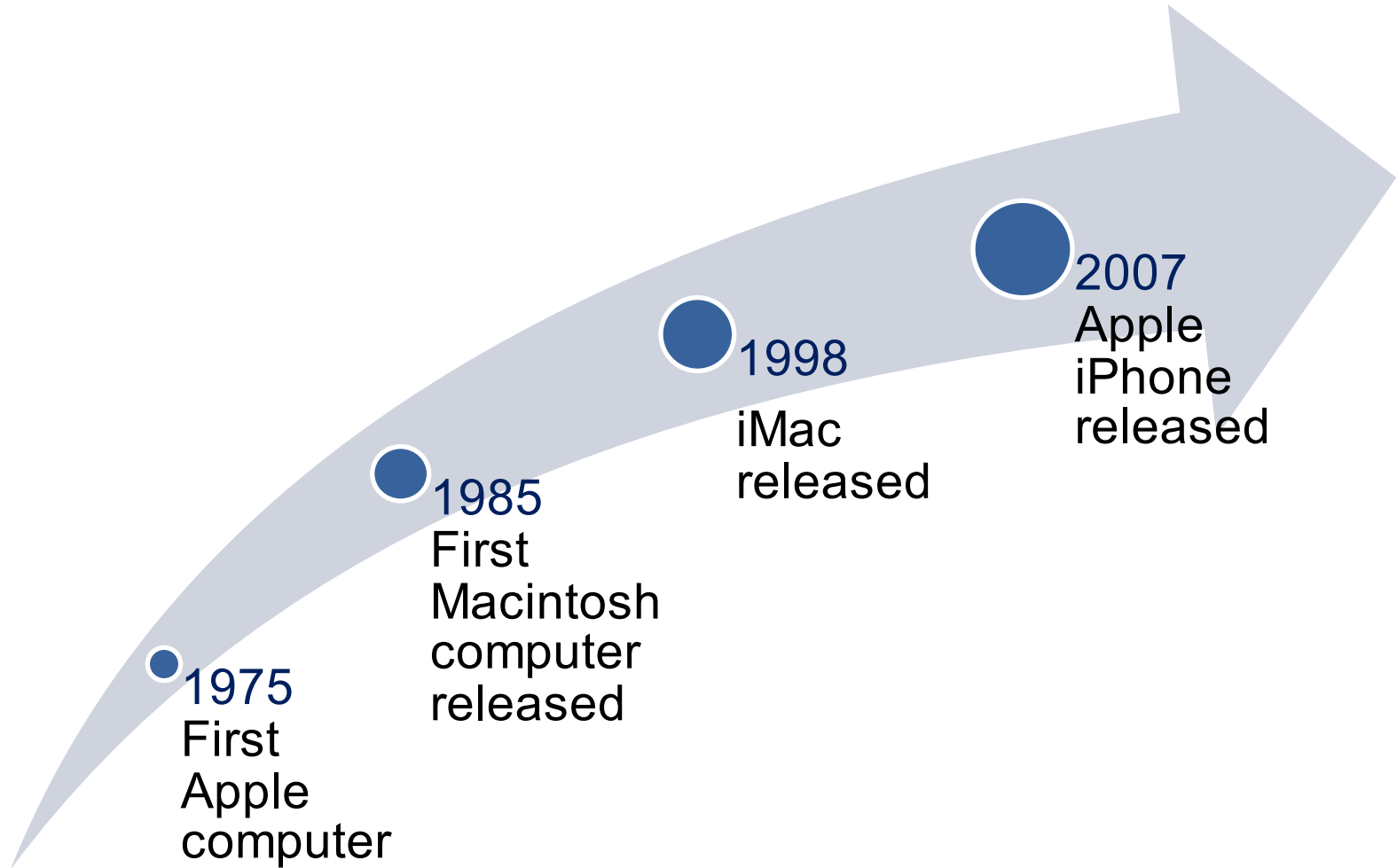
Learning Objective

- Summarize various types of digital forensics.

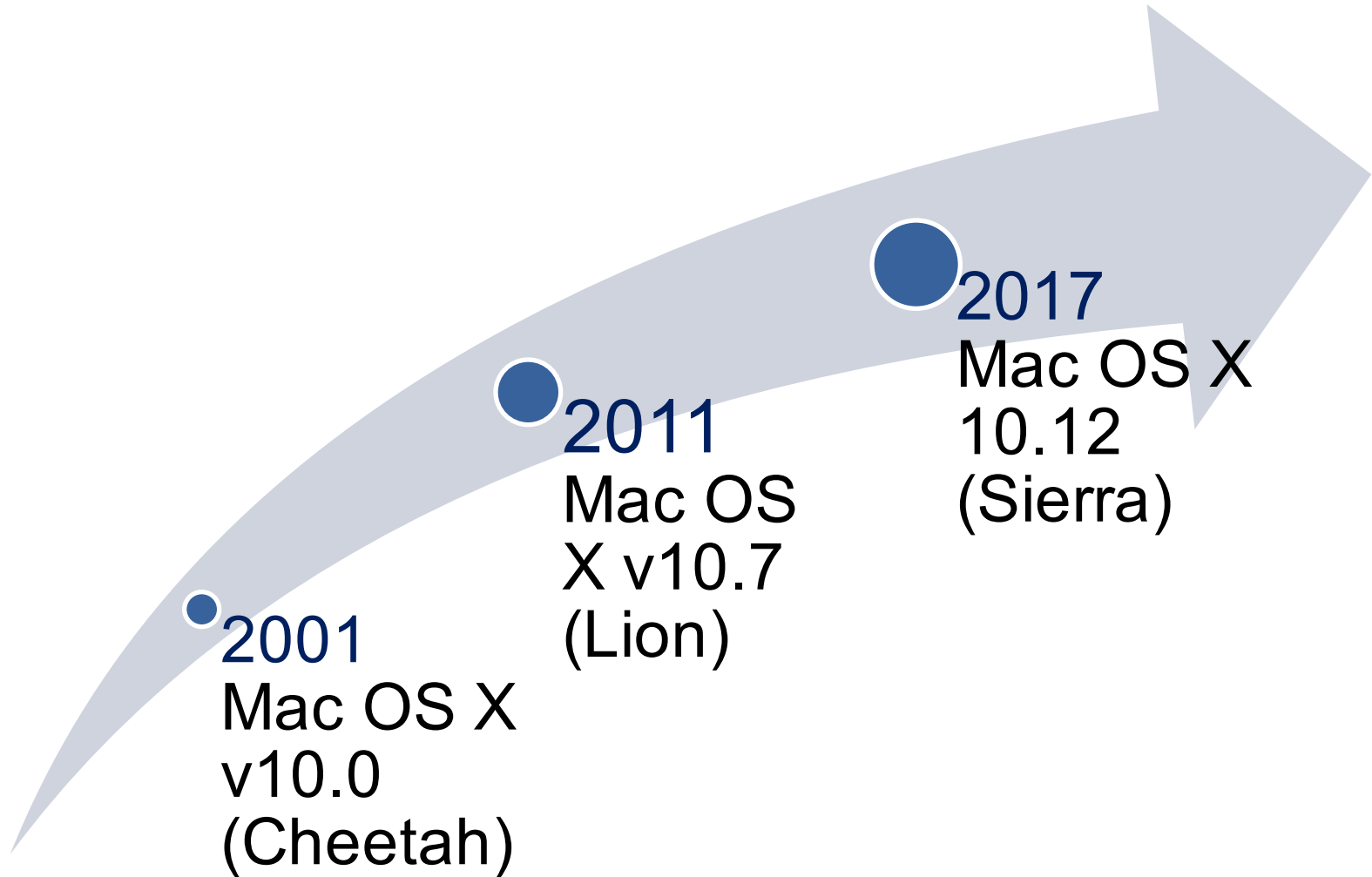
Key Concepts

- Macintosh basics
- Where to find the logs in Macintosh
- Forensically interesting Macintosh directories
- Forensic techniques for Macintosh
- How to undelete files in Macintosh

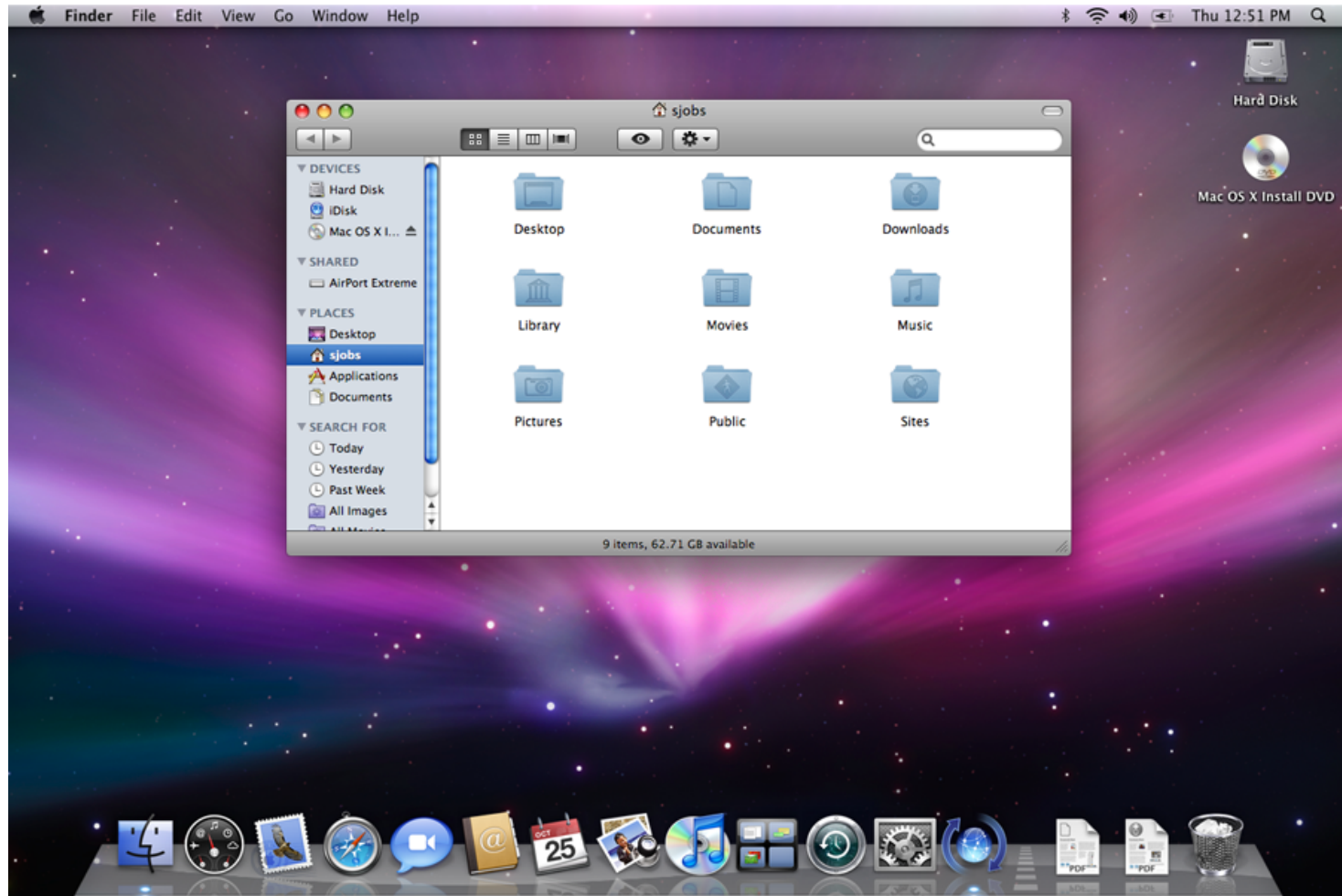
History of Apple/Macintosh



History of Mac OS X



Mac OS X Desktop



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Macintosh File Systems

Macintosh File System (MFS)

- Older technology no longer in use

Hierarchical File System (HFS)

- Used on Macintosh Plus
- Beginning with Mac OS 8.1, replaced by HFS+

Hierarchical File System Plus (HFS+)

- Enhancement of HFS
- Preferred file system on Mac OS X

Hierarchical File System Plus (HFS+)

- Supports journaling
- Supports disk quotas
- Has hard and soft links
- Uses 32 bits for allocation blocks
- Supports long filenames, up to 255 characters
- Uses Unicode

Hierarchical File System Plus (HFS+) and Forensics

- Supports aliases
- Performs defragmentation on a per-file basis

Supported File Systems

ISO9660 (extensions)

Microsoft Disk Operating System (MS-DOS)

New Technology File System (NTFS)

Universal Disk Format (UDF)

UNIX File System (UFS)

Partition Schemes

GUID Partition Table

Apple Partition Map

Master Boot Record (MBR)

Macintosh Logs

`/var/log/`

- General log repository
- Data on removable media and serial numbers

`/var/spool/cups`

- Information about printed documents
- Name of document, user who printed

`/Library/Receipts`

- Information about system and software updates

Macintosh Logs (Cont.)

`/Users/<user>/
.bash_history`

- Information on Bash shell activity

`/var/vm`

- Contains subfolder named app profile
- Lists of recently opened applications, temporary data used by applications

`/Users/`

- User files and preferences

Directories

- Macintosh has a number of important directories
- Some are relevant to a forensic examination of a Macintosh machine

Volumes

- Contains information about mounted devices
- Includes data regarding:
 - Hard disks
 - External disks
 - CDs
 - Digital video discs (DVDs)
 - Virtual machines (VMs)

/Users, /Applications, /Network, and /etc

/Users

User accounts and
associated files

/Applications

All applications

/Network

Information about
servers, network
libraries, and
network properties

/etc

Configuration files

/Library/Preferences/System Configuration/dom.apple.preferences. plist

- Contains the network configuration data for each network card

Macintosh Forensic Techniques

Target Disk mode

Searching virtual memory

Shell commands

Target Disk Mode

- Create a forensically sound copy of disk contents
 - `dd` and `netcat`
 - Imaging tools within EnCase or Forensic Toolkit
- Begin in Target Disk Mode
 - Cannot write to disk
 - No chance of altering source disk
 - Connect to the suspect computer with via USB or FireWire and image the disk

Target Disk Mode (Cont.)



Screenshot republished with permission from Apple Inc.

Searching Virtual Memory

- Swap file/virtual memory is in `/var/vm/`
- Check it with Linux commands
 - `ls` returns list of files
 - `ls -al` returns list of all files in virtual memory, who launched program and when
 - `grep` lets you search in virtual memory folder

Using Shell Commands

`date`

- Returns current date and time zone

`ls
/dev/disk?`

- Lists current device files in use

`/hdiutil
partition
/dev/disk0`

- Lists the partition table for the boot drive

Using Shell Commands (Cont.)

`system_
profiler
SPHardware
DataType`

- Returns hardware information for host system

`system_
profiler
SPSoftware
DataType`

- Returns information about the operating system

How to Examine a Mac

- Many forensics tools are less effective in extracting data on a Macintosh than in Windows.
- One technique is to create a copy of the forensic image and then mount it as a read-only virtual machine (VM).
 - Mount it as read only.

Deleting Files on Macintosh

- Similar to Windows, when file is deleted, references to file are gone and clusters might be used and overwritten
- Even if data is overwritten, data might exist in unallocated space and in index nodes
- Deleted files moved to the trash folder, similar to Recycle Bin in Windows
- Macintosh trash folder is .Trash, a hidden folder on the root directory of file system

Undeleting Files on Macintosh (Cont.)

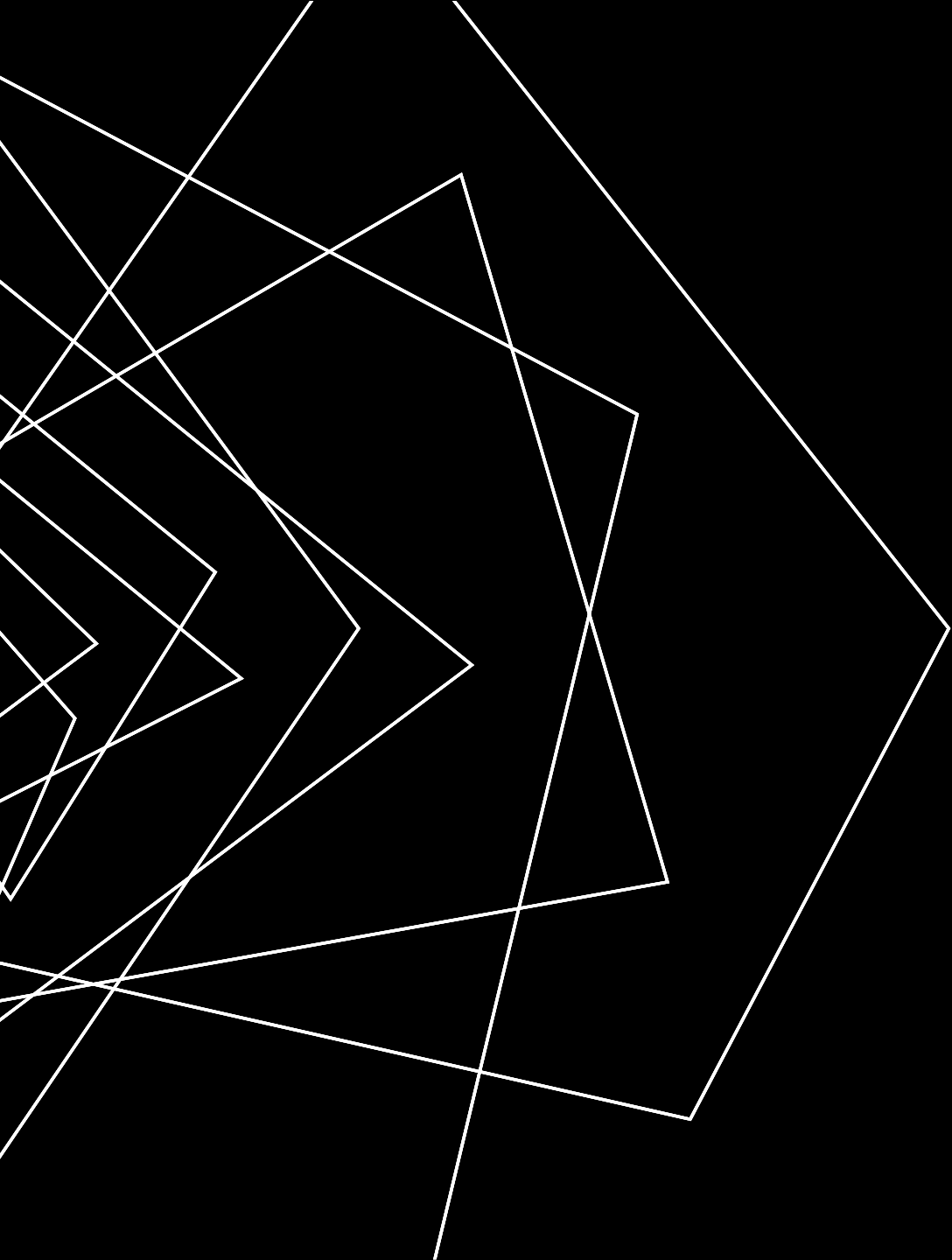
- Recover deleted files from .Trash by copying or moving to other location
or
- Use tools to recover files, even after trash bin has been emptied
 - Mac Undelete
 - Free Undelete
 - MacKeeper

Listing Contents of .Trash

```
$/Trash ls -al
total 764
drwx----- 7 pc pc 306 Oct 30 15:05 .
drwxr-xr-x 30 pc pc 1054 Oct 30 12:44 ..
-rw----- 1 pc pc 6148 Oct 30 14:38 .DS_Store
-rw-r--r-- 1 pc pc 187500 Oct 27 15:41 Resume.pdf
-rw-r--r-- 1 pc pc 108382 Oct 27 15:43 VacationPIC.jpg
-rw-r--r-- 1 pc pc 108382 Oct 27 15:43 Report.pdf
```

Summary

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THANK YOU

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